

6 Intramolecular electronic transitions

In this lecture, we will look more carefully at how molecules change electronic states, either because the transition is induced by a light field or because of nonadiabatic coupling. First we want to understand under what circumstances transitions may take place. We will look at the problem from a time-independent and time-dependent perspective.

6.1 Fermi's Golden-rule

The so-called Fermi golden rule goes back to Wentzel and Dirac (~ 1927) but was named after Fermi's lecture notes (~ 1950).

Consider a state which can decay by (for example) a nonadiabatic transition into a continuum or quasi-continuum. This is the physical situation which occurs in predissociation. Now the Hamiltonian matrix is

$$\mathbf{H} = \begin{pmatrix} E_1 & \Delta_{12} & \Delta_{13} & \cdots \\ \Delta_{12} & E_2 & \Delta_{23} & \cdots \\ \vdots & \vdots & \vdots & \ddots \end{pmatrix}. \quad (6.1)$$

If we start in state m , $b_m = 1$ but $b_n = 0$ for all $n \neq m$. We will use first-order perturbation theory and thus make the approximation that Δ_{nm} is small such that higher-order perturbation terms are not important. This gives

$$b_n(t) = \frac{-1}{i\hbar} \int_0^t e^{-iE_n(t-t')/\hbar} \Delta_{nm} e^{-iE_m t'/\hbar} dt' \quad (6.2)$$

$$= \frac{-\Delta_{nm}}{i\hbar} e^{-iE_n t/\hbar} \int_0^t e^{i\Omega_{nm} t'} dt' \quad (6.3)$$

$$= \frac{-\Delta_{nm}}{i\hbar} e^{-iE_n t/\hbar} \left[\frac{e^{i\Omega_{nm} t'}}{i\Omega_{nm}} \right]_0^t \quad (6.4)$$

$$= \frac{\Delta_{nm}}{\hbar} e^{-iE_n t/\hbar} \frac{e^{i\Omega_{nm} t} - 1}{\Omega_{nm}}, \quad (6.5)$$

where $\Omega_{nm} = (E_n - E_m)/\hbar$.

Therefore the population of state n at time t is

$$P_{m \rightarrow n}(t) = |b_n(t)|^2 = \frac{\Delta_{nm}^2}{\hbar^2} \frac{4 \sin^2(\Omega_{nm} t/2)}{\Omega_{nm}^2}. \quad (6.6)$$

At short times,

$$\lim_{t \rightarrow 0} P_{m \rightarrow n}(t) = \frac{\Delta_{nm}^2}{\hbar^2} t^2 \quad (6.7)$$

the population increases quadratically with t . This is the same as the behaviour we saw in tunnelling in the ammonia molecule. It is a sign of *coherent* dynamics.

If we are interested not in the rate of a particular transition but of any transition away from m into any state n , we can compute the time-dependent population lost from state m as

$$P_{m \rightarrow ?} = \sum_n P_{m \rightarrow n}(t) \quad (6.8)$$

This function is particularly interesting to study in the long-time limit. In this limit, $\frac{4 \sin^2(\Omega_{nm}t/2)}{\Omega_{nm}^2}$ tends to a delta function $2\pi t \delta(\Omega_{nm}) = 2\pi \hbar t \delta(E_n - E_m)$ to give

$$\lim_{t \rightarrow \infty} P_{m \rightarrow ?}(t) = \frac{2\pi}{\hbar} t \sum_n \Delta_{nm}^2 \delta(E_n - E_m). \quad (6.9)$$

Therefore, in the long-time limit, the population increases linearly with time. Thus a transition is only possible if the final state has the same energy as the initial state. Energy is required to be conserved in such transitions. It is only valid if the first-order perturbation theory is appropriate. This would not be the case if you had a two-state system as then, in the long-time limit, one would need to take higher-order terms into account in order to obtain the oscillatory dynamics we saw previously. However, if there are a huge number of possible final states, each one only increases its population very slightly, and thus the first-order perturbation is valid and hence taken together the population transition occurs linearly in time. This is a sign of *incoherent* dynamics. You can easily see for yourself how the dynamics changes completely in these two regimes using the simple Bixon-Jortner model and starting in a reactant state which is coupled to a large number of product states uncoupled to each other. Plotting the population of the reactant state against time will show incoherent dynamics if the density of product states is high enough and coherent dynamics if the density is low enough. In intermediate regimes, it gets complicated and there is no simple phenomenological description.

We can define the rate constant for transitions out of m :

$$k_m = \lim_{t \rightarrow \infty} P_{m \rightarrow ?}(t)/t \quad (6.10)$$

$$= \sum_n \frac{2\pi}{\hbar} \Delta_{nm}^2 \delta(E_n - E_m). \quad (6.11)$$

To find the rate in thermal equilibrium, we simply sum over the rates from initial states m weighted by the Boltzmann distribution

$$k(T) = \sum_m e^{-\beta E_m} \sum_n \frac{2\pi}{\hbar} \Delta_{nm}^2 \delta(E_n - E_m). \quad (6.12)$$

6.2 Floquet theory

In the previous section, we saw that transitions only occur if the energy of the initial and final states match. At first sight this does not appear to be the case for electronic absorption spectroscopy, in which a ground-to-excited state transition occurs. Floquet theory provides a useful way to think about the interaction between molecules and light, which will help us understand how these transitions occur within the Golden-rule picture.

If the Hamiltonian is time-dependent and *periodic*, there is a special solution of the time-dependent Schrödinger equation:

$$i\hbar \frac{\partial}{\partial t} |\Psi(t)\rangle = \hat{H}(t) |\Psi(t)\rangle. \quad (6.13)$$

We insert the Ansatz $|\Psi(t)\rangle = e^{-iE_\lambda t/\hbar} |\Phi_\lambda(t)\rangle$ to obtain

$$E_\lambda e^{-iE_\lambda t/\hbar} |\Phi_\lambda(t)\rangle + i\hbar e^{-iE_\lambda t/\hbar} \frac{\partial}{\partial t} |\Phi_\lambda(t)\rangle = \hat{H}(t) e^{-iE_\lambda t/\hbar} |\Phi_\lambda(t)\rangle \quad (6.14)$$

which can be rearranged to give

$$\hat{H}(t) |\Phi_\lambda(t)\rangle - i\hbar \frac{\partial}{\partial t} |\Phi_\lambda(t)\rangle = E_\lambda |\Phi_\lambda(t)\rangle. \quad (6.15)$$

This defines the Floquet eigenvalues and eigenstates. The most general solution of the TDSE is thus $|\Psi(t)\rangle = \sum_\lambda a_\lambda e^{-iE_\lambda t/\hbar} |\Phi_\lambda(t)\rangle$.

The Floquet theorem states that if the Hamiltonian is periodic, then so are the functions $|\Phi_\lambda(t)\rangle$ which solve the previous equation. Therefore we can expand the Hamiltonian and wavefunction as a Fourier series:

$$\hat{H}(t) = \sum_{n=-\infty}^{\infty} e^{in\omega t} \hat{H}^{(n)} \quad (6.16)$$

$$|\Phi_\lambda(t)\rangle = \sum_{n=-\infty}^{\infty} e^{in\omega t} |\phi_\lambda^{(n)}\rangle \quad (6.17)$$

where the period is $2\pi/\omega$.

$$\sum_{n'} e^{in'\omega t} \hat{H}^{(n')} \sum_n e^{in\omega t} |\phi_\lambda^{(n)}\rangle - i\hbar \sum_n in\omega e^{in\omega t} |\phi_\lambda^{(n)}\rangle = E_\lambda \sum_n e^{in\omega t} |\phi_\lambda^{(n)}\rangle \quad (6.18)$$

multiply both sides by $e^{-im\omega t}$ and average time over one period

$$\sum_{n,n'} e^{i(n'+n-m)\omega t} \hat{H}^{(n')} |\phi_\lambda^{(n)}\rangle - i\hbar \sum_n in\omega e^{i(n-m)\omega t} |\phi_\lambda^{(n)}\rangle = E_\lambda \sum_n e^{i(n-m)\omega t} |\phi_\lambda^{(n)}\rangle \quad (6.19)$$

$$\sum_{n,n'} \delta_{n',m-n} \hat{H}^{(n')} |\phi_\lambda^{(n)}\rangle - i\hbar \sum_n in\omega \delta_{nm} |\phi_\lambda^{(n)}\rangle = E_\lambda \sum_n \delta_{nm} |\phi_\lambda^{(n)}\rangle \quad (6.20)$$

$$\sum_n \hat{H}^{(m-n)} |\phi_\lambda^{(n)}\rangle - m\hbar\omega |\phi_\lambda^{(m)}\rangle = E_\lambda |\phi_\lambda^{(m)}\rangle \quad (6.21)$$

In the typical case where $\hat{H}(t) = \hat{H}_0 - 2\varepsilon\mu \cos(\omega t)$, we have that $\hat{H}^{(0)} = \hat{H}_0$ (the molecule-only Hamiltonian) and $\hat{H}^{(1)} = \hat{H}^{(-1)} = -\varepsilon\mu$ (interaction with the light field) whereas all other $\hat{H}^{(n)}$ terms are 0. This gives rise to the following effective *time-independent* Hamiltonian

$$\begin{pmatrix} \hat{H}_0 + 2\hbar\omega & -\varepsilon\mu & 0 & 0 & 0 \\ -\varepsilon\mu & \hat{H}_0 + \hbar\omega & -\varepsilon\mu & 0 & 0 \\ 0 & -\varepsilon\mu & \hat{H}_0 & -\varepsilon\mu & 0 \\ 0 & 0 & -\varepsilon\mu & \hat{H}_0 - \hbar\omega & -\varepsilon\mu \\ 0 & 0 & 0 & -\varepsilon\mu & \hat{H}_0 - 2\hbar\omega \end{pmatrix}$$

of which only a small section is shown as it is infinite in size.

Without actually going into quantum-field theory,* we have effectively included quantized photons of energy $\hbar\omega$. In this way we can understand absorption/emission processes as nonadiabatic coupling between molecular/light states and Fermi's golden-rule can be used to compute the probability of transition from one state to another. For instance absorption is described by the dynamical transformation from $|\phi^{(n)}\rangle$ to $|\phi^{(n-1)}\rangle$ via the coupling $-\varepsilon\mu$. This process starts in a state where the molecule and n photons are present (total Hamiltonian $\hat{H}_0 + n\hbar\omega$) and ends in a state where the molecule and $n - 1$ photons are present (total Hamiltonian $\hat{H}_0 + (n - 1)\hbar\omega$). Therefore, to conserve energy we require that the internal energy of the molecule increases by $\hbar\omega$.

6.3 Preparation of an excited electronic state

Now instead of monochromatic light, we will use a pulsed field (laser pulse) with carrier frequency ω

$$\varepsilon(t) = \varepsilon f(t) \cos(\omega t) \quad (6.22)$$

where the normalized function $f(t)$ describes the pulse shape and ε is the amplitude. The simplest pulse shape is the Gaussian:

$$f(t) = \sqrt{\frac{1}{2\pi\tau_p^2}} \exp\left\{-\frac{(t - t_p)^2}{2\tau_p^2}\right\}. \quad (6.23)$$

which is centred at t_p and has width (pulse duration) τ_p .

The Hamiltonian is no longer periodic, so we return to the more general time-dependent picture.

Generalizing equation (5.49) we get

$$|\chi_e^{(1)}(t)\rangle = \frac{-\varepsilon}{i\hbar} \int_{t_0}^t e^{-i\hat{H}_e(t-t')/\hbar} \hat{\mu}_{eg} |\chi_g(t_0)\rangle f(t') \cos(\omega t') e^{-i\omega_g(t'-t_0)} dt'. \quad (6.24)$$

We analyse this in the basis of vibrational eigenstates of the excited electronic state, $|\chi_{en}\rangle$, with eigenvalues $\hbar\omega_{en}$:

$$\langle\chi_{en}|\chi_e^{(1)}(t)\rangle = \frac{-\varepsilon}{i\hbar} \int_{t_0}^t \langle\chi_{en}|e^{-i\hat{H}_e(t-t')/\hbar} \hat{\mu}_{eg} |\chi_g(t_0)\rangle f(t') \cos(\omega t') e^{-i\omega_g(t'-t_0)} dt' \quad (6.25)$$

$$= \frac{-\varepsilon}{i\hbar} \int_{t_0}^t e^{-i\omega_{en}(t-t')} \langle\chi_{en}|\hat{\mu}_{eg} |\chi_g(t_0)\rangle f(t') \cos(\omega t') e^{-i\omega_g(t'-t_0)} dt'. \quad (6.26)$$

Therefore the amplitude for populating a given vibrational eigenstate n is determined by two things, the Frank-Condon overlap, $\langle\chi_{en}|\hat{\mu}_{eg} |\chi_g(t_0)\rangle$, and the Fourier transform of the pulse:

$$\tilde{f}(\Omega_n) = \int f(t') e^{i\Omega_n t'} dt' = e^{-\Omega_n^2 \tau_p^2 / 2 + i\Omega_n \tau_p} \quad (6.27)$$

where $\Omega_n = \omega_{en} - \omega_g - \omega$. Therefore for long pulses, only states corresponding to $\Omega_n \approx 0$ can be occupied, but for short pulses, many states can be occupied. This is another example of the uncertainty principle: the pulse is either localized in time or energy.

*If you do want to find out more about this fascinating topic, I recommend the book Tom Lancaster and Stephen J. Blundell *Quantum field theory for the gifted amateur* (Oxford).

In previous lectures we have treated the limit of long pulse duration $\tau_p \rightarrow \infty$, from which one recovers the continuous wave limit and absorption spectrum. Here we take the limit of short pulse duration, $\tau_p \rightarrow 0$, which is equivalent to $f(t) \rightarrow \delta(t - t_p)$.

$$|\chi_e^{(1)}(t)\rangle = \frac{-\varepsilon}{i\hbar} e^{-i\hat{H}_e(t-t_p)/\hbar} \hat{\mu}_{eg} |\chi_g(t_0)\rangle \cos(\omega t_p) e^{-i\omega_g(t_p-t_0)} \quad (6.28)$$

In this limit, the wavepacket is effectively just placed on the excited state at $t = t_p$.

6.4 Photochemistry

Consider a system of three electronic states. For simplicity, we employ the diabatic representation.

$$\hat{H}(t) = \hat{H}_0 + \hat{H}_1(t) \quad (6.29)$$

$$= \frac{\hat{p}^2}{2m} + \begin{pmatrix} \hat{V}_0 & 0 & \hat{V}_{02} \\ 0 & \hat{V}_{11} & \hat{V}_{12} \\ \hat{V}_{20} & \hat{V}_{21} & \hat{V}_{22} \end{pmatrix} + \begin{pmatrix} 0 & -\mu_{10}\varepsilon(t) & 0 \\ -\mu_{01}\varepsilon(t) & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \quad (6.30)$$

The kinetic energy term is often written like this, but a (3×3) -identity matrix is implied such that it only adds to the diagonal elements of $\hat{H}_0$. Here we have assumed that the light field couples states 0 to 1, but that state 2 is “dark” and has no transition dipole moment coupling to the ground state, perhaps because of symmetry reasons. We have also assumed that there is no diabatic coupling between states 0 and 1, but allowed there to be some between 2 and 0. This may simply be because there is no crossing between V_0 and V_1 .

A laser pulse may prepare a wavepacket in state 1 which will evolve in the coupled set of diabatic states. It may cross over to state 2 and from there find its way back to state 0. Let’s see how we can describe this using time-dependent perturbation theory and propagators.

The initial state is $|\psi(t_0)\rangle = |\phi_0\rangle |\chi_0(t_0)\rangle$ and the dipole operator is $\hat{\mu}_{10} |\phi_1\rangle \langle \phi_0|$. Taking the limit of a short pulse,

$$|\psi^{(1)}(t)\rangle = \frac{-\varepsilon}{i\hbar} e^{-i\hat{H}_0(t-t_p)/\hbar} \hat{\mu}_{10} |\phi_1\rangle \langle \phi_0 | \psi(t_0) \rangle \cos(\omega t_p) e^{-i\omega_g(t_p-t_0)} \quad (6.31)$$

The interesting dynamics is thus described by

$$e^{-i\hat{H}_0(t-t_p)/\hbar} |\psi^{(1)}(t_p)\rangle \quad (6.32)$$

where

$$|\psi^{(1)}(t_p)\rangle = |\phi_1\rangle |\chi_1^{(1)}(t_p)\rangle = \frac{-\varepsilon}{i\hbar} \hat{\mu}_{10} |\phi_1\rangle |\chi_0(t_0)\rangle \cos(\omega t_p) e^{-i\omega_g(t_p-t_0)} \quad (6.33)$$

We write this as a vector of nuclear wavefunctions in the basis of electronic states:

$$|\psi^{(1)}(t_0)\rangle = \begin{pmatrix} 0 \\ |\chi_1^{(1)}(t_p)\rangle \\ 0 \end{pmatrix} \quad (6.34)$$

Now we use the Trotter splitting on the propagator

$$\hat{U}(\epsilon) = e^{-i\hat{H}_0\epsilon/\hbar} = e^{-i\hat{V}\epsilon/2\hbar} e^{-i\hat{T}\epsilon/\hbar} e^{-i\hat{V}\epsilon/2\hbar} + \mathcal{O}(\epsilon^3) \quad (6.35)$$

The only difference now is that each of these operators is a (3×3) -matrix in the basis of electronic states.

For instance,

$$\hat{A} = e^{-i\hat{V}\epsilon/\hbar} = \begin{pmatrix} \hat{A}_{00} & \hat{A}_{01} & \hat{A}_{02} \\ \hat{A}_{10} & \hat{A}_{11} & \hat{A}_{12} \\ \hat{A}_{20} & \hat{A}_{21} & \hat{A}_{22} \end{pmatrix} \quad (6.36)$$

and

$$\hat{B} = e^{-i\hat{T}\epsilon/\hbar} = \begin{pmatrix} e^{-i\hat{T}\epsilon/\hbar} & 0 & 0 \\ 0 & e^{-i\hat{T}\epsilon/\hbar} & 0 \\ 0 & 0 & e^{-i\hat{T}\epsilon/\hbar} \end{pmatrix} \quad (6.37)$$

such that $\hat{U}(\epsilon) = \hat{A}\hat{B}\hat{A}$ to give

$$\begin{pmatrix} |\chi_0^{(1)}(t+\epsilon)\rangle \\ |\chi_1^{(1)}(t+\epsilon)\rangle \\ |\chi_2^{(1)}(t+\epsilon)\rangle \end{pmatrix} = \begin{pmatrix} \hat{U}_{00}(\epsilon) & \hat{U}_{01}(\epsilon) & \hat{U}_{02}(\epsilon) \\ \hat{U}_{10}(\epsilon) & \hat{U}_{11}(\epsilon) & \hat{U}_{12}(\epsilon) \\ \hat{U}_{20}(\epsilon) & \hat{U}_{21}(\epsilon) & \hat{U}_{22}(\epsilon) \end{pmatrix} \begin{pmatrix} |\chi_0^{(1)}(t)\rangle \\ |\chi_1^{(1)}(t)\rangle \\ |\chi_2^{(1)}(t)\rangle \end{pmatrix} \quad (6.38)$$

6.5 Further Reading

The following textbooks will give you further information on the topics of this lecture:

Fermi's golden rule

- Zwanzig, Nonequilibrium statistical mechanics (Oxford), section 3.1

Bixon-Jortner Model

- Heller, The Semiclassical Way to Dynamics and Spectroscopy, chapter 14

Floquet theory

- Tannor, Introduction to quantum mechanics: a time-dependent perspective, section 9.5

Preparation of an excited electronic state

- May and Kühn, Charge and Energy Transfer Dynamics in Molecular Systems (Wiley, 2011), section 6.5
<https://onlinelibrary.wiley.com/doi/book/10.1002/9783527633791>
 (full access through ETH library – use a VPN)

Hopefully you may be able to read them on the internet.